

A Forecast Model For Crypto Currencies

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Abstract

This paper examines whether the well known equity **momentum** effect exists in cryptocurrency markets. We estimate daily cross-sectional regressions of forward returns on past returns with daily price data from Binance for about 50 most actively traded cryptocurrencies. The resulting time series of regression coefficients serves as a measure of the momentum effect's strength and evolution over time. With the results, we propose a dollar-neutral trading strategy that goes long the top quartile and short the bottom quartile of cryptocurrencies ranked by their momentum signal. Our historical simulation shows the strategy has better sharpe ratio and drawdown profile than simply buying and holding Bitcoin alone.

1 Introduction

Cryptocurrencies have become increasingly popular among younger generations over the past decade, and the trend shows no signs of slowing down. Several factors help explain this growing adoption.

- Crypto offers accessibility — anyone with a smartphone and an internet connection can participate in the market, with no need for a brokerage account or minimum balance.
- Younger investors are drawn to the asymmetric upside potential; early Bitcoin and Ethereum adopters became millionaires, and stories like these spread quickly through social media.
- Crypto resonates culturally with a generation that grew up distrusting traditional financial institutions, particularly in the wake of the 2008 financial crisis. Decentralization and the ethos of financial self-sovereignty appeal to those skeptical of banks and governments.
- The gamification of trading platforms and the rise of meme coins have made crypto feel more like entertainment than investing, lowering the psychological barrier to entry.
- Finally, crypto markets never close — they trade 24 hours a day, seven days a week — which suits a younger, always-online demographic that doesn't want to wait for Wall Street's opening bell.

As a result, millions of retail investors have poured into crypto trading, many of them with limited financial experience. The appeal is obvious: the potential for outsized returns in a short period of time is difficult to find elsewhere. But the crypto market is notoriously volatile, and that volatility cuts both ways. Prices can double in a matter of weeks and collapse just as quickly. This was on full display recently when Bitcoin touched an all-time high of approximately \$109,000 in January 2025 before suffering a dramatic drawdown, falling over 30% in the weeks that followed. For leveraged retail traders or those without a disciplined risk management framework, moves like this can be catastrophic.

This is precisely the challenge we are trying to address. Rather than simply buying and holding, or making directional bets on individual coins, we are exploring a more systematic and risk-aware approach to trading cryptocurrencies. In related work, Chen [1] attempts to forecast Bitcoin returns using a set of macroeconomic indicators. While the results show some promising predictive signal, the strategy's simulation performance remains highly sensitive to volatile market conditions, and the overall risk-adjusted returns fall short of what would be required for practical investment deployment. In contrast, this paper shifts the focus from predicting the absolute returns of a single cryptocurrency to predicting the relative returns across a cross-section of cryptocurrencies. By targeting relative rather than directional performance, the dollar-neutral portfolio construction naturally hedges out broad market moves, reducing

the strategy’s sensitivity to the violent swings that characterize the crypto market as a whole and that proved challenging for single-asset forecasting approaches.

In the early 1990s, Fama and French [2,3] published a series of groundbreaking papers that fundamentally reshaped how academics and practitioners think about systematic patterns in stock market returns. Prior to their work, the Capital Asset Pricing Model (CAPM) dominated financial theory, positing that a single factor — market beta — was sufficient to explain the cross-sectional variation in expected returns. Fama and French challenged this view directly, demonstrating empirically that two additional characteristics carried significant and independent explanatory power:

- **Size**, measured by market capitalization
- **Value**, measured by the ratio of book value to market value.

Small-cap stocks and high book-to-market stocks had historically earned returns that could not be explained by their market beta alone, suggesting the presence of systematic risk premia that CAPM had missed. Shortly after, Jegadeesh and Titman [4] documented another powerful anomaly that would become equally influential: **momentum**. They showed that stocks which had performed well in the past tended to continue outperforming over the subsequent months, while past losers tended to continue underperforming. This cross-sectional momentum effect was strong, persistent, and difficult to explain away as a compensation for risk under standard asset pricing models. Carhart [5] later incorporated momentum as a fourth factor alongside the original Fama-French three, cementing its status as one of the most robust and widely studied phenomena in empirical finance. The standard construction uses a 12-month lookback window, skipping the most recent month to avoid short-term reversal contamination. Of particular interest here is the momentum factor. The intuition is straightforward: stocks that have outperformed their peers over the past year tend to carry positive investor sentiment, improving earnings revisions, and institutional buying momentum that persists for months before eventually mean-reverting. The effect has since been documented across asset classes — equities, bonds, commodities, and currencies — suggesting it reflects something fundamental about how markets process information and how investor behavior generates predictable mispricings. This raises a natural question: do similar patterns exist in cryptocurrency markets? Crypto is in many ways a very different arena — it is dominated by retail rather than institutional investors, operates continuously without market close, lacks the earnings fundamentals that anchor equity valuation, and is subject to sentiment and narrative-driven price swings of a magnitude rarely seen in traditional markets. And yet precisely because of these characteristics, one might expect behavioral biases such as trend-chasing, herding, and delayed information diffusion to be even more pronounced, potentially making momentum effects stronger rather than weaker. In this paper, we investigated whether the cross-sectional momentum anomaly has a counterpart in the cryptocurrency market, and whether it can be systematically exploited through a disciplined, dollar-neutral trading strategy.

2 Data

We constructed the dataset by downloading daily OHLCV (Open, High, Low, Close, Volume) price data for approximately thirty of the most actively traded cryptocurrencies from the Binance public API, which provides free access to historical market data without requiring authentication. For each cryptocurrency, we retrieved the longest available history up to ten years, recognizing that many altcoins have existed for considerably less time than Bitcoin and Ethereum — some having launched only within the past few years. The resulting panel is therefore unbalanced, with Bitcoin and Ethereum having the longest histories and newer coins entering the sample at their respective listing dates. This unbalanced structure is handled naturally in our cross-sectional framework, as each daily regression simply uses whatever coins have sufficient price history available on that date. Daily close-to-close returns are then computed as

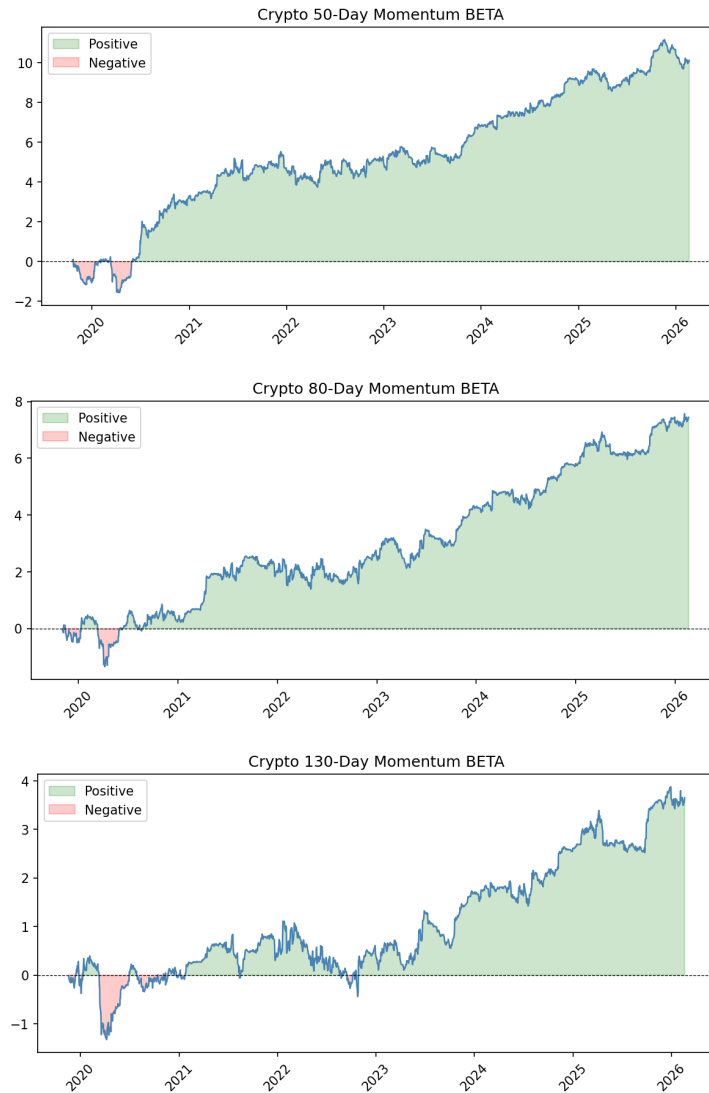
$$r_t = \ln(p_t/p_{t-1}) \tag{1}$$

Since cryptocurrency markets operate continuously — 24 hours a day, seven days a week — we aligned the daily price series to the NYSE trading calendar, forward-filling weekend and holiday prices to ensure consistency with the broader financial literature and to facilitate comparison with equity-based momentum strategies.

3 Momentum Effects

To estimate the momentum effect, we run a daily cross-sectional regression of each coin's forward one-day return on its normalized lagged returns with various lookbacks. Before running the regression, the lagged return of each coin is standardized into a z -score by dividing by its one year rolling standard deviation, which ensures that coins with different levels of price volatility are placed on a comparable scale. Both the z -scored past returns and the forward returns are then subtracted by the cross-sectional average across all coins, so that the regression captures purely relative performance rather than directional moves of the overall crypto market. The slope from each daily regression, or β , measures the strength of the momentum effect on that particular day. Repeating this procedure for every trading day in the sample yields a time series of daily β s.

Our results reveal that momentum effects are statistically significant across multiple lookback horizons. Specifically, we find strong and consistent momentum signals using 50-day, 80-day, and 130-day lagged returns as the ranking variable, suggesting that the predictability of relative cryptocurrency returns is not an artifact of a particular parameter choice but rather a robust feature of the data. The strength of the momentum effect exhibits a modest decline as the lookback window lengthens, with the 50-day horizon producing the strongest and most consistent signal, followed by the 80-day and 130-day horizons respectively. This pattern suggests that shorter-term relative performance is a somewhat more reliable predictor of future returns in cryptocurrency markets, potentially reflecting the faster rate at which information is incorporated into prices in a market dominated by retail participants and sentiment-driven trading. Nevertheless, the effect remains statistically meaningful across all three horizons, which motivates our decision to combine them rather than rely on any single lookback window alone.



4 Trading Strategy

Having established the presence of momentum effects across multiple horizons, we turn to the construction of a trading strategy. Rather than committing to a single lookback window, we combine all three momentum signals with the goal of producing a more stable and robust forecast through diversification across formation periods. A natural approach would be to combine the three signals into a single forecast with a linear model. However, since the three momentum signals are highly correlated with one another by construction, ordinary least squares regression is prone to numerical instability and unreliable coefficient estimates due to multicollinearity. An additional practical consideration in favor of the equal-weighted combination is the limited length of our sample relative to comparable studies in the literature. A linear model combining the three momentum signals would require a dedicated in-sample period for parameter estimation, leaving a shorter out-of-sample window for strategy evaluation. Given that our dataset is already relatively brief, reserving a portion of the data for model fitting would meaningfully reduce the power of the out-of-sample simulation. We therefore adopt a simpler and more transparent approach: treating each of the three signals as an equal vote and defining the final forecast as their unweighted average. This ensemble approach implicitly applies equal shrinkage to each signal, avoids the overfitting and collinearity issues inherent in multivariate regression, and has the additional practical virtue of introducing no free parameters that need to be estimated from historical data. The equal-weighted combination of correlated predictors is well-supported in the forecasting literature, where simple averages have repeatedly been shown to match or outperform more complex combination schemes out of sample.

For each momentum signal, to obtain a stable and forward-looking estimate of β for portfolio construction, we compute a rolling average of the daily β s using a two-year trailing window. This rolling average smooths out day-to-day noise in the regression coefficients and provides a more reliable estimate of the prevailing momentum effect. The expected return of each coin on day t is then computed as the product of its demeaned z -scored lagged return and the previous day's rolling average β , ensuring no lookahead bias is introduced.

These expected returns serve as the ranking signal for portfolio construction. Each day, we sort all available coins by their expected returns and long the top quartile and short the bottom quartile, with equal dollar weighting within each side. The long book is sized at \$1,000 and the short book at -\$1,000, resulting in a dollar-neutral portfolio with zero net market exposure. The daily profit and loss is then computed as the dot product of the portfolio weights and the realized forward returns. Repeating this construction every trading day over the full sample period yields a historical simulation of the strategy's performance across a wide range of market conditions.

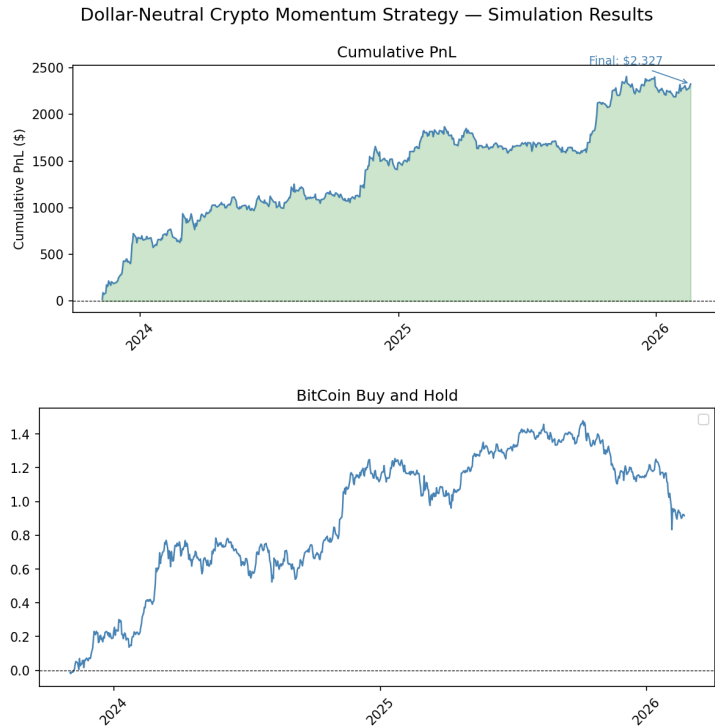


Table 1: Strategy Performance vs. Bitcoin Buy-and-Hold

Metric	Momentum Strategy	Bitcoin Buy-and-Hold
Sharpe Ratio (365-day)	1.98	0.82
Total Return	116.3%	90.7%
Max Drawdown	-14.5%	-49.7%

The momentum strategy delivers a substantially superior risk-adjusted performance compared to a simple Bitcoin buy-and-hold over the evaluation period. The strategy achieves an annualised Sharpe ratio of 1.98, more than double that of Bitcoin at 0.82, indicating that the returns generated per unit of risk are significantly higher. The maximum drawdown of the momentum strategy is approximately 14.5% of gross exposure, a fraction of Bitcoin’s peak-to-trough decline of 49.7%, highlighting the risk-reduction benefit of the dollar-neutral construction. These results suggest that systematically exploiting cross-sectional momentum across a diversified basket of cryptocurrencies offers a more disciplined and capital-efficient approach to gaining exposure to the crypto market than outright directional buying.

5 Conclusion

This study investigates the presence of cross-sectional momentum in cryptocurrency markets and demonstrates that it can be systematically exploited through a simple and transparent trading strategy. Motivated by the well-established momentum factor in the equity literature, we show that past relative returns over three different lookback horizons are significant predictors of future relative returns across a universe of fifty actively traded cryptocurrencies. By combining the three momentum signals through equal-weighted averaging, we mitigate the collinearity and overfitting concerns that would arise from a multivariate regression approach, while benefiting from the diversification across formation periods. The resulting dollar-neutral portfolio strategy, which goes long the top quartile and short the bottom quartile of coins ranked by their momentum signal, generates an annualised Sharpe ratio of 1.98 over the simulation period — more than double that of a Bitcoin buy-and-hold benchmark — while limiting the maximum drawdown to approximately 14.5% of gross exposure compared to 49.7% for Bitcoin. These findings suggest that the momentum anomaly documented in traditional financial markets extends meaningfully to the cryptocurrency space, and that systematic cross-sectional strategies offer a viable and risk-aware alternative to the directional speculation that dominates retail crypto trading. Future work could explore the application of additional risk factors, transaction cost modelling, and the extension of the universe to a broader set of digital assets.

References

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